Investment Demand Shock and Business Cycle

A Bayesian estimation of a DSGE model

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Motivation

- **Justiniano, Primiceri and Tambalotti (2010):** an estimated DSGE attributes a large share of U.S. fluctuations to a **investment-specific technology (IST) shock** ($K_{t+1} = (1-\delta)K_t + \mu_{s,t} I_t$).
- **The comovement puzzle:** IST shocks struggle to reproduce the joint positive comovement of consumption, investment, and inflation seen in the data.
- **Nirei and Ragot (2025):** proposed an **investment demand shock**, which can generate this comovement, but lacks empirical test.

This paper

Estimate a DSGE model with both an IST shock and an investment demand shock to quantify how important the investment demand shock is for the U.S. business cycle.

Investment demand shock and its origin

Reduced-form **investment demand (ID) shock**:

$$I_t = \mu_{d,t} I_t^*, \qquad \ln \mu_{d,t} = (1-\rho_d) \ln \mu_d + \rho_d \ln \mu_{d,t-1} + \varepsilon_{d,t}.$$

A toy New-Keynesian model with **lumpy investment assumption**,

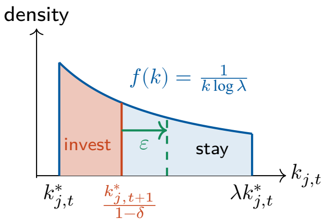
$$i_{j,t} \in \{0,\ (\lambda-1)(1-\delta)k_{j,t}\}.$$

- Invest iff future capital below a threshold,

$$k^*_{j,t+1} = a^{\eta-1}_{j,t+1} \Phi_t Y_{t+1}.$$

- Decision rule in terms of current capital:

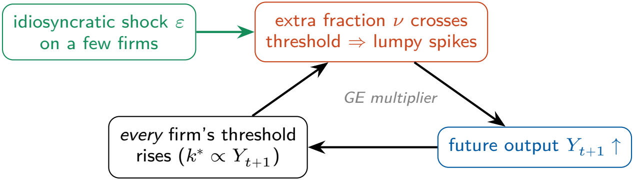
$$i_{j,t} = \begin{cases} (\lambda-1)(1-\delta)k_{j,t}, & k_{j,t} < \frac{k^*_{j,t+1}}{1-\delta} \\ 0, & \text{otherwise} \end{cases}$$



Idiosyncratic technology shock ε increases threshold:

$$k^*_{j,t+1}(\varepsilon) = (\varepsilon a_{j,t+1})^{\eta-1} \Phi_t Y_{t+1}$$

Why idiosyncratic shocks do not wash out: a GE multiplier



$$\text{Total induced inv.} = \sum_{i=1}^{\infty} \text{induced fraction}^{(i)} \times \text{ average induced inv.}^{(i)}$$

- The induced fraction and average induced investing do not decay due to the lumpy investment assumption.
- With many firms, the idiosyncratic risk itself averages out, but the multiplied response does not.
  $\Longrightarrow$ Aggregate investment behaves as if hit by a common shock:

$$I_t = \mu_{d,t}\, I_t^*.$$

The estimated DSGE model

Based on Smets and Wouters (2007), Herbst and Schorfheide (2015), augmented with the investment demand shock.

- **Nominal rigidities:** Calvo price setting; Erceg nominal wage rigidity.
- **Other frictions:** investment adjustment cost and endogenous capital utilization.
- **Policy:** Taylor rule; output-stabilizing government spending.
- **Five structural shocks:** technology ε_a, IST ε_s, monetary ε_R, government ε_g, and investment demand ε_d.

Estimation: method and data

Method

- Bayesian estimation; methodology follows Smets and Wouters (2007), Justiniano et al. (2010).
  - Posterior via Metropolis–Hastings: $100{,}000$ draws.

Data

- Six quarterly U.S. observables (FRED), **1983:Q1–2002:Q4**:
  - output growth, CPI inflation, fed funds rate,
  - hours, investment growth, gov.-spending growth.

Baseline result: variance decomposition

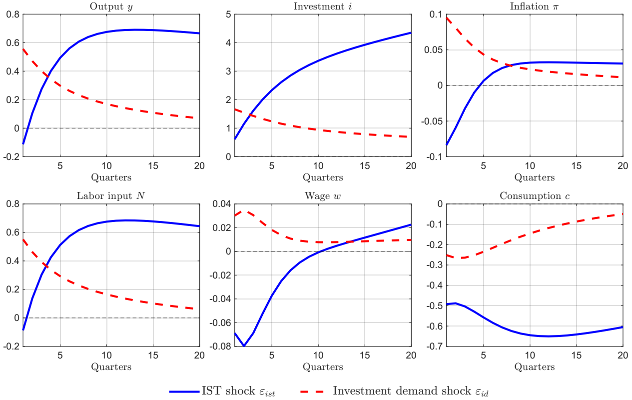
ID shock alone explains $\approx \mathbf{29}\%$ of output and $\approx \mathbf{38}\%$ of investment FEVD.

Observable	ε_g	ε_s	ε_R	ε_a	ε_d	meas
Output	2.48	31.77	3.40	46.81	**19.97**	1.38
Inflation	0.32	12.21	26.73	47.43	11.13	1.62
Interest rate	0.42	18.55	14.76	48.89	14.30	4.52
Hours	3.73	49.06	1.70	14.27	**26.52**	8.24
Investment	0.45	55.80	0.26	19.66	**24.67**	1.37
Gov. spending	95.93	0.07	0.01	0.09	0.07	3.79

Frequency-domain FEVD (%), business-cycle band **6–32** quarters. The decomposition is obtained using the spectrum of the DSGE model and an inverse first difference filter for output, consumption abd investment to reconstruct the levels.

ε_g gov., ε_s IST, ε_R monetary, ε_a technology, ε_d ID.

Baseline result: impulse responses



IST shock ⇒ deflationary

- **Intertemporal substitution**: cheap investment ⇒ save more today.
- labor supply shift dominates ⇒ Wage ↓ while hours ↑.

ID shock ⇒ inflationary

- **Wealth effect**: higher future output raises lifetime wealth ⇒ less impactful substitution effect.
- labor demand shift dominates ⇒ Wage ↑ and hours ↑.

Robustness: Smets–Wouters-style model

- Add habit formation, preference / price- / wage-markup shocks, and consumption and wage growth as observables.
- **ID shock turns marginal:** only 4.4% of output and 13.6% of investment FEVD.

Observable	ε_g	ε_s	ε_R	ε_a	ε_b	ε_p	ε_w	ε_d
Output	6.15	10.31	7.86	43.58	17.21	5.00	5.52	**4.37**
Inflation	0.05	1.78	0.82	18.85	0.40	53.05	25.04	0.00
Interest rate	0.19	7.16	30.37	26.67	4.51	10.39	20.72	0.00
Hours	2.00	20.43	7.10	34.85	7.86	1.10	26.35	0.30
Investment	0.25	54.98	2.43	18.59	3.19	0.62	6.33	**13.62**
Wage	0.00	0.12	0.14	18.25	0.30	16.96	64.23	0.00
Consumption	0.25	2.36	6.48	43.49	42.01	1.92	3.49	0.00
Gov. spending	99.90	0.01	0.01	0.05	0.02	0.01	0.01	0.00

Unconditional FEVD (%): Smets–Wouters-style model. ε_g gov., ε_s IST, ε_R monetary, ε_a technology, ε_b preference, ε_p price-markup, ε_w wage-markup, ε_d ID.

Takeaway

- Revisits investment-side shocks by testing **investment demand shock** in an estimated DSGE model.
- **Baseline:** the ID shock alone explains $\approx 40\%$ of output and $\approx 50\%$ of investment FEVD—a competitive driver—and generates an inflationary expansion with comovement (vs. deflationary IST).
- **Limitations:**
  - results are specification-sensitive: the SW-style model shrinks the ID shock.
  - no direct data identifies the ID shock—identification leans on comovement differences, hard to test over the whole parameter space;
  - imperfect fit and some extreme posteriors in the baseline;

Backup: toy model details

Specifying the investment demand shock

The reduced-form investment demand shock $\mu_{d,t}$:

$$I_t = \mu_{d,t}\, I_t^*, \quad \ln\mu_{d,t} = (1-\rho_d)\ln\mu_d + \rho_d \ln\mu_{d,t-1} + \varepsilon_{d,t}.$$

- **Step 1:** Optimal problem with i_t

$$\begin{aligned} V^{pre}(b_t, k_t, A_t) = \max_{c_t, b_{t+1}, i_t} & \; u(c_t) + \beta\mathbb{E}_t\left(V^{pre}(b_{t+1}, k_{t+1}, A_{t+1})\right) \\ s.t. \quad & c_t + b_{t+1} + i_t = R_t(A_t)b_t + r_t^k(A_t)k_t \\ & k_{t+1} = (1-\delta)k_t + i_t \end{aligned}$$

- Denote the investment choice of the first optimal problem by i_t^*.
- **Step 2**: Given i_t^*

$$\begin{aligned} V^{post}(b_t, k_t, A_t, \mu_{d,t}) = \max_{c_t, b_{t+1}} & \; u(c_t) + \beta\mathbb{E}_t\left(V^{pre}(b_{t+1}, k_{t+1}, A_{t+1})\right) \\ s.t. \quad & c_t + b_{t+1} + i_t^*\mu_{d,t} = R_t(A_t)b_t + r_t^k(A_t)k_t \\ & k_{t+1} = (1-\delta)k_t + i_t^*\mu_{d,t} \end{aligned}$$

From idiosyncratic shock to investment demand shock

The reduced-form investment demand shock $\mu_{d,t}$:

$$I_t = \mu_{d,t}\, I_t^*, \quad \ln \mu_{d,t} = (1-\rho_d)\ln \mu_d + \rho_d \ln \mu_{d,t-1} + \varepsilon_{d,t}.$$

Intuition: Firms decide their investment according to the its own **productivity** and **aggregate output**; (i) if some firms receive positive productivity shock, they will increase their investment; (ii) the increase in investment means more future output; (iii) thus firms not influenced by the shock will also invest more.

Standard New-Keynesian CES aggregator and demand; $a_{j,t} \in \{a_L, a_H\}$ is deterministic:

$$Y_t = \left(\int_0^1 y_{j,t}^{\frac{\eta-1}{\eta}}\, dj\right)^{\frac{\eta}{\eta-1}}, \quad y_{j,t} = \left(\frac{p_{j,t}}{P_t}\right)^{-\eta} Y_t, \quad y_{j,t} = a_{j,t} k_{j,t}.$$

Combining demand and production, real revenue is

$$r_{j,t}(a_{j,t}, k_{j,t}) = \frac{p_{j,t} y_{j,t}}{P_t} = \left(a_{j,t} k_{j,t}\right)^{1-\frac{1}{\eta}}\, Y_t^{\frac{1}{\eta}}.$$

- Concave in own capital ($\eta > 1$): diminishing returns through the demand curve.

Lumpy investment and the investment threshold

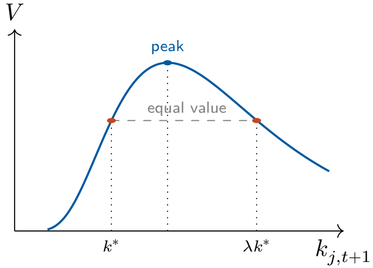
Firms solve

$$\underbrace{\max_{\{i_{t+\tau}\}} \sum_{\tau\geq 0} \Lambda_{t,t+\tau}(r_{j,t+\tau} - i_{j,t+\tau}),}_{:=V}$$

s.t. $k_{j,t+\tau+1} = (1-\delta)k_{j,t+\tau} + i_{j,t+\tau}$,

$i_{j,t+\tau} \in \{0,\ (\lambda-1)(1-\delta)k_{j,t+\tau}\}$,

$\lambda > 1$.



Indifference at k^* pins down the threshold:

$$k^*_{j,t+1} = a_{j,t+1}^{\eta-1}\,\Phi_t(\Lambda_t, \Lambda_{t+1})\,Y_{t+1}$$

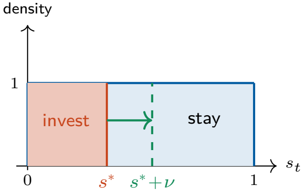
Normalize a firm's position by its **distance to the threshold**

$$s_{j,t+1} = \frac{\log(k_{j,t+1}/k^*_{j,t+1})}{\log\lambda} \implies i_{j,t} = \begin{cases}(\lambda-1)(1-\delta)k_{j,t}, & s_{j,t+1} < 0\\ 0, & s_{j,t+1} \geq 0\end{cases}$$

(i): Idiosyncratic shock $\to$ investing fraction

Without investing, under stationary distribution $(Y = Y_t = Y_{t+1},\ \Phi = \Phi_t = \Phi_{t+1})$, $s_{j,t+1}$ drifts:

$$s_{j,t+1} = s_{j,t} - \underbrace{\frac{-\log(1-\delta) + (\eta - 1)\log\left(a_{t+1}/a_t\right)}{\log\lambda}}_{:=s^*(a_t,a_{t+1})}$$



Assume $s_t \mid a_t \sim U[0,1)$ under stationary distribution.

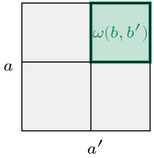
- Then $s^*(a_t, a_{t+1}) \times 1$ is the **stationary investing fraction**
- Add an $i.i.d.$ shock ε on a_{t+1} for the firms in group $(a_t = b, a_{t+1} = b')$:

$$s^*(b, \varepsilon b') = s^*(b, b') + \underbrace{\frac{(\eta - 1)\log\varepsilon}{\log\lambda}}_{:=\nu}.$$

- The cutoff in that group rises by ν, so an extra $\nu \times 1$ fraction invests.

(ii)&(iii): Investing fraction → aggregate output → investing fraction

More investment in (b, b') raises **future output** of this group, which is a part of future aggregate output:



$$Y_{t+1}(b, b', \nu) = \Big(\omega(b, b') \int_0^{s^*+\nu} \big(b'\lambda(1-\delta)k_t(s_t)\big)^{\frac{\eta-1}{\eta}} \, dF(s_t \mid b) + \text{no investment firms}\Big)^{\frac{\eta}{\eta-1}}$$

- Higher Y_{t+1} shifts the cutoff in every group, and induce an extra investing fraction:

$$\nu^{(1)}(a, a') = \frac{\log(Y_{t+1}(\nu)/Y_{t+1}(0))}{\log \lambda}, \quad \nu^{(1)} := \sum_{(a,a')} \omega(a, a') \, \nu^{(1)}(a, a').$$

- The first ν of investing firms raises Y, which induces $\nu^{(1)}$ more, which raises Y again ... a GE multiplier that iterates to a new equilibrium.

Total induced investment $\gg \nu$

$$\text{Total induced inv.} = \sum_{i=1}^{\infty} \text{induced fraction}^{(i)} \times \text{ average induced inv.}^{(i)}$$

- **Scale of induced fraction**

$$\lim_{\nu \to 0} \frac{\nu^{(1)}}{\omega(b, b')\, \nu} = 1$$

The $\nu^{(1)}$ has the **same scale** as the original fraction
$\implies$ each $\omega(a, a')\nu^{(1)}(a, a')$ will induce the same scale $\nu^{(2)}(\nu^{(1)}(a, a'))$ again.

- **Scale of average induced investment**

$$\left.\frac{\partial \Delta x(a, a', \nu)}{\partial \nu}\right|_{\nu=0} = (\lambda - 1)(1 - \delta)\lambda^{s^*} a^{\eta - 1} \Psi Y$$

The derivative says $\Delta x(a, a', \nu) = O(\nu)$.
$\implies$ average induced investment will not decay faster than ν.